John Garang

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SUMMARY

Software engineer with 2+ years of experience in backend and web development, working mainly in Java, Spring Boot and relational databases, and building RESTful APIs and React interfaces. Currently pursuing an MSc in Data Science and Business Analytics at Bocconi, where I apply machine learning and NLP to real datasets.

EXPERIENCE

Freelance Web Developer **Jan 2024 – Present**
Remote

- Developed frontend applications using HTML, CSS, JavaScript and ReactJS, creating reusable components that reduced development time for subsequent projects.
- Built and deployed RESTful APIs enabling communication between frontend and backend systems.
- Worked directly with non-technical clients to translate business requirements into technical solutions, with clear documentation.

Software Development Intern **Jan 2023 – Sep 2023**
Eclectics International *Nairobi, Kenya*

- Configured a backend system using Spring Boot and Spring Security, supporting reliable user operations and system performance.
- Maintained data integrity and storage using Spring JPA and MySQL, ensuring secure handling of files.
- Diagnosed and resolved technical issues, reducing downtime and improved application stability.

EDUCATION

MSc in Data Science and Business Analytics **Sep 2025 – Jul 2027**
Bocconi University, Milan, Italy
Coursework: Statistics & Probability, Machine Learning, Natural Language Processing, Programming & Databases

BSc in Information Technology (Final Grade: 3.5/4.00) **Jan 2020 – Jun 2024**
Jomo Kenyatta University of Agriculture and Technology, Nairobi, Kenya
Coursework: Web Development (PHP, Python, Java, JavaScript)

PROJECTS

Healthy Food Alternative Intelligence from Textual Ingredient Data – *NLP, Bocconi*

- Built an NLP pipeline in Python that predicts NOVA food-processing level from product ingredient text, combined with topic modelling and healthier-alternative recommendation. A tuned Bag-of-Words Logistic Regression reached 93.03% accuracy and 93.04% macro F1 on the test set.

Food Recipe Provenance – *Machine Learning, Bocconi*

- Classified recipes as American or Italian from 40 TF-IDF ingredient features using a two-level stacking ensemble (LightGBM, XGBoost, Extra Trees, RBF-SVM with a Logistic Regression meta-learner), achieving about 92% 5-fold cross-validation accuracy.

Credibility Without Purchases – *Simulation and Modeling, Bocconi*

- Agent-based network model of 90 banks, showing how the ECB's 2012 OMT announcement compressed Italian and Spanish sovereign spreads without any bond purchases.

SCHOLARSHIPS, COMPETITIONS & CERTIFICATIONS

- Bocconi Scholarship
- DAFI Scholarship
- Winning Group of VARGROUP Hackathon in Rimini, Italy
- Fundamentals of Data Science in Precision Medicine and Cloud Computing – Stanford Medicine
- Software Engineering: Backend and Frontend – ALX Africa | Web Developer Bootcamp – Udemy

TECHNICAL SKILLS

Backend: Spring Framework, Django, Flask, Python, RESTful APIs, Agentic AI
Frontend: ReactJS, HTML, CSS, JavaScript
Databases: PostgreSQL, MySQL, MongoDB
Spoken Languages: English (Proficient), Dinka (Native), Swahili (Proficient), Italian (Intermediate, B1)
Interests: Football, Basketball, Artificial Intelligence, Natural Language Processing, Machine Learning